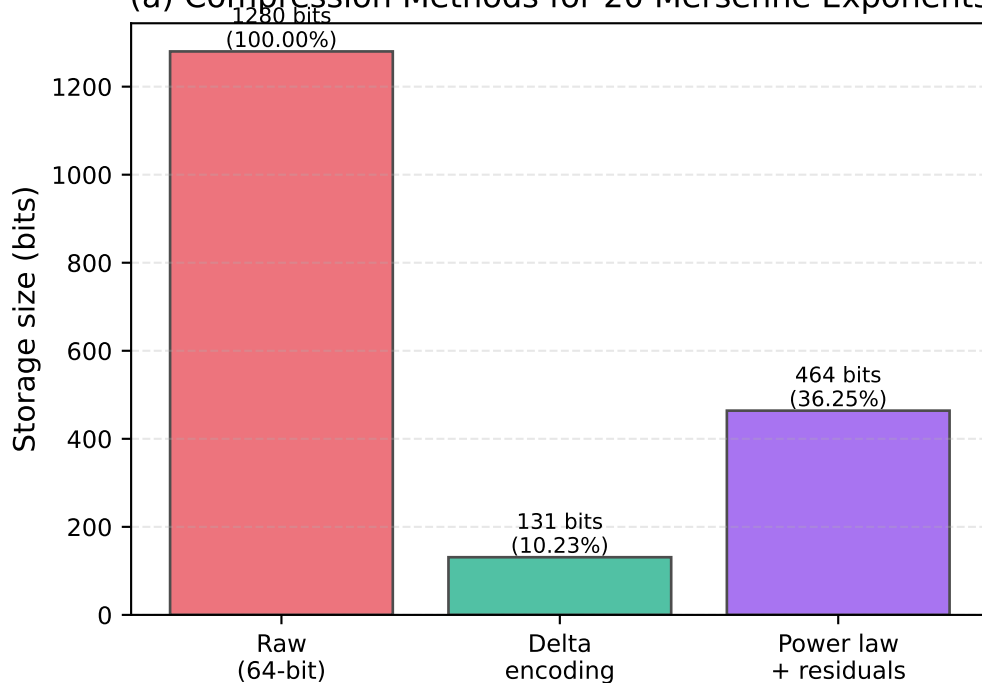


(a) Compression Methods for 20 Mersenne Exponents



(b) Infinite Compression Theorem

Perfect numbers can be compressed from infinitely many $\sim 2^p$ bit representations to:

1. Sequence of exponents $\{p_n\}$
2. Euclid-Euler formula: $N = 2^{p-1}(2^p - 1)$

Further compression via:

- Power-law scaling: $p_n \sim n^{4.32}$
- Dimensional structure: $D \approx 4.56$
- Binary pattern: p ones + $(p - 1)$ zeros